

ch25 Radiation and Radioactivity

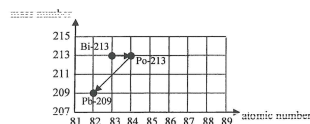
Items:

2026 Q12

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12. (a) Bismuth-213 (Bi-213) is a radionuclide currently undergoing clinical trials for cancer treatment. It is prepared in liquid and injected into the human body, where it targets and destroys cancer cells. Figure 12.1 shows a decay series of Bi-213 .

Figure 12.1



- (i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

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- *(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11} . Find the activity (in Bq) at $t = 0$ s. (2 marks)

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- *(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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- (b) In hospital, technicians wear protective gloves to handle liquid radioactive materials, such as those containing Bi-213 . To monitor radiation exposure, they wear a specialised ring on their finger, as shown in Figure 12.2(a). Each ring contains a small chip in a plastic enclosure (Figure 12.2(b)), which detects and records radiation doses. The rings are collected every three months to monitor the radiation exposure of the technicians.



Figure 12.2(a)

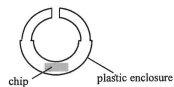


Figure 12.2(b)

- (i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand? (1 mark)

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- (ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

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- (iii) Which type(s) of radiation (α , β or γ) can reach the chip of the ring? (1 mark)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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